

IET CLN smart city

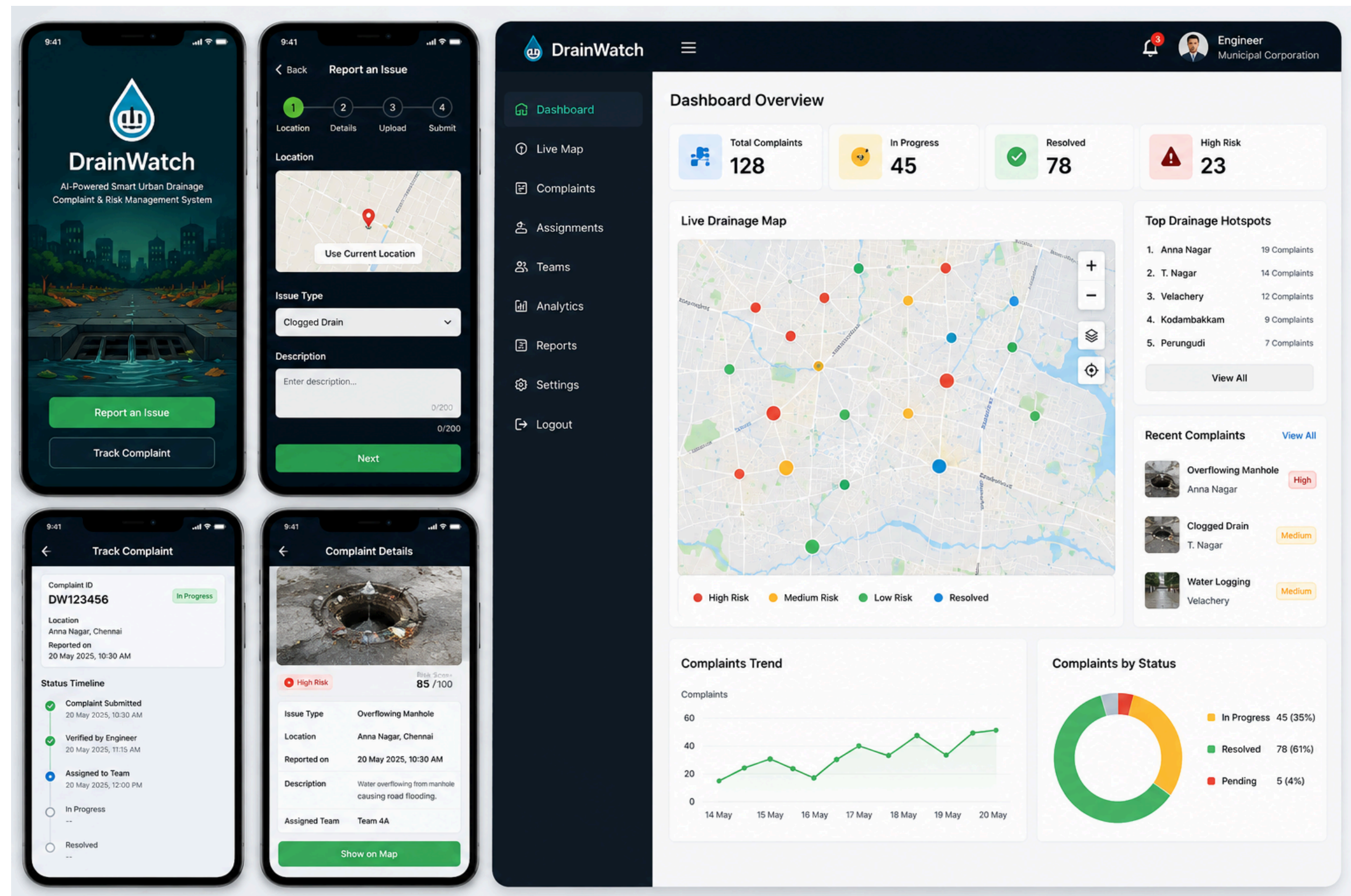
To develop an AI-powered Smart Drainage Management System for real-time issue reporting, intelligent prioritization, and proactive flood prevention.

Team Name
ZeroBug

Team Members
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Objective

To develop an AI-powered smart drainage management system for real-time issue reporting, intelligent prioritization, and proactive flood prevention.



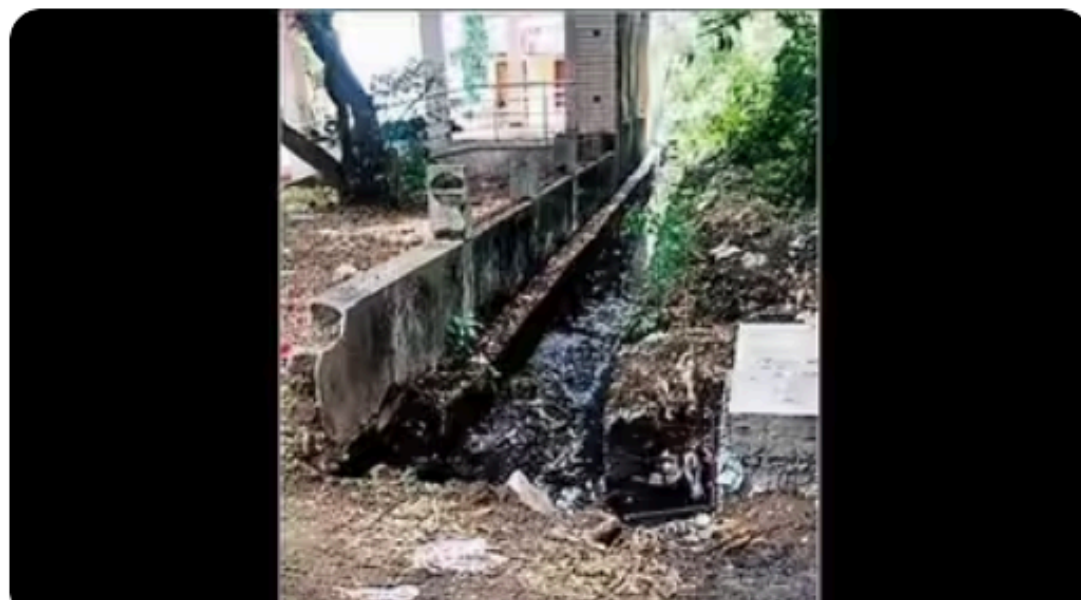
Motivation

Why Do We Need DrainWatch?

- Urban flooding has become a recurring problem in Indian cities due to clogged drains, poor maintenance, and rapid urbanization.
- Manual complaint systems delay issue reporting and resolution.
- Municipal authorities lack real-time visibility into drainage hotspots.
- AI-based prioritization can help resolve high-risk drainage issues before they become disasters.

Chromepet grapples with flooding, drainage issues

May 23, 2025, 04:32 IST



Chennai: Illegal discharge of sewage from residential and commercial establishments into storm water drains is causing repeated flooding and sewage overflows in Chromepet. Lack of proper maintenance of these drains especially in areas such as GST Road and Mummoorthy Nagar adds to the problem as it causes severe stagnation and inundation during the monsoon.

Rainwater stagnates on GST Road between Chromepet, Pallavaram

The shops on GST Road were among the worst affected, as dirty water stagnated at entrances, disrupting business and forcing customers to wade through the sludge



Key Facts

- India has announced a \$300 million urban flood mitigation program for major cities, including Chennai, recognizing the growing need for better drainage management.
- Rapid concretization is reducing natural water absorption, making cities more vulnerable to flooding during heavy rains.
- Even short periods of rainfall now cause severe waterlogging in many Chennai localities because drains are blocked or poorly maintained.

DrainWatch addresses these challenges by combining AI, GIS mapping, and citizen reporting to enable faster complaint resolution, proactive maintenance, and smarter urban drainage management.

Chennai rains: Drainage woes out in the open

Omjasvin M D / TNN / Updated: Dec 13, 2024, 12:07 IST

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Chennai experienced widespread waterlogging in over 135 localities following 10cm of rainfall over 36 hours. Knee-deep inundation affected areas like Velachery and Vyasarpadi, disrupting traffic and daily life. Blocked storm water drains were the primary cause, while some areas benefited from retention walls.



CHENNAI: About 10cm rain spread across 36 hours is not heavy by any stretch of imagination, and yet at least 135 localities in the city suffered water-logging with residents wading through knee-deep water on Thursday. While office-goers and businesses suffered due to strong morning-hour spells, low-lying localities like Velachery, Vyasarpadi, Kodungaiyur, and Pulianthope faced up to knee-deep inundation. As Pazhavanthangal and Aranganathan subways were inundated, access to GST Road and Anna Salai was cut off for several hours, leading to traffic snarls.

Garbage we dump everywhere clogs drains, floods homes in Chennai

Padmaja J / TNN / Updated: Oct 17, 2024, 08:39 IST

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Residents in Chennai waded through polluted floodwaters filled with garbage, while the Greater Chennai Corporation utilized drones and excavators to address the issue. Blocked canals and clogged waterbodies exacerbated flooding. Authorities emphasize imposing fines for garbage dumping and monitoring ca ... [Read More](#)



Garbage floats in stagnant rainwater at Pattalam

CHENNAI: In areas that experienced significant flooding from Tuesday, residents had to wade through not only floodwaters but also plastic bags, old furniture, and other waste. In Pattalam and Vyasarpadi flood waters pushed garbage into homes. Greater Chennai Corporation deployed drones to monitor garbage dumping and excavators to pick them up from channels supposed to drain residential areas. But garbage still blocked some canals and even got washed downstream.

Problem Definition

Existing Problem

- Drainage complaints are reported through manual or fragmented systems, causing delays.
- Authorities lack real-time monitoring and accurate location information.
- No intelligent system exists to prioritize high-risk drainage issues.
- Blocked drains and overflowing manholes lead to urban flooding, traffic disruption, and public health risks.
- Maintenance is reactive rather than proactive, resulting in inefficient resource utilization.

Current drainage management is manual, slow, and lacks AI-driven monitoring, leading to delayed responses and increased urban flood risks.

Plastic waste puts millions of world's poorest at higher risk from floods

More than 200 million face more intense and frequent floods due to plastic pollution blocking drainage systems, report finds



📷 People walk on a flooded street after torrential rains paralysed Mumbai in July 2005.
Photograph: Sebastian D'Souza/AFP/Getty Images

Proposed Solution

DrainWatch

AI-Powered Smart Urban Drainage Complaint & Risk Management System

Our Solution

- Citizens report drainage issues using a mobile/web app.
- Upload photos with GPS location.
- AI analyzes the complaint and assigns a risk score.
- GIS maps display complaints and identify hotspots.
- Municipal engineers receive prioritized complaints.
- Citizens receive live status updates until resolution.

'DRAINWATCH - Urban Drainage Management Workflow



© DrainWatch - AI-Powered Smart Urban Drainage Complaint & Risk Management System

Innovation

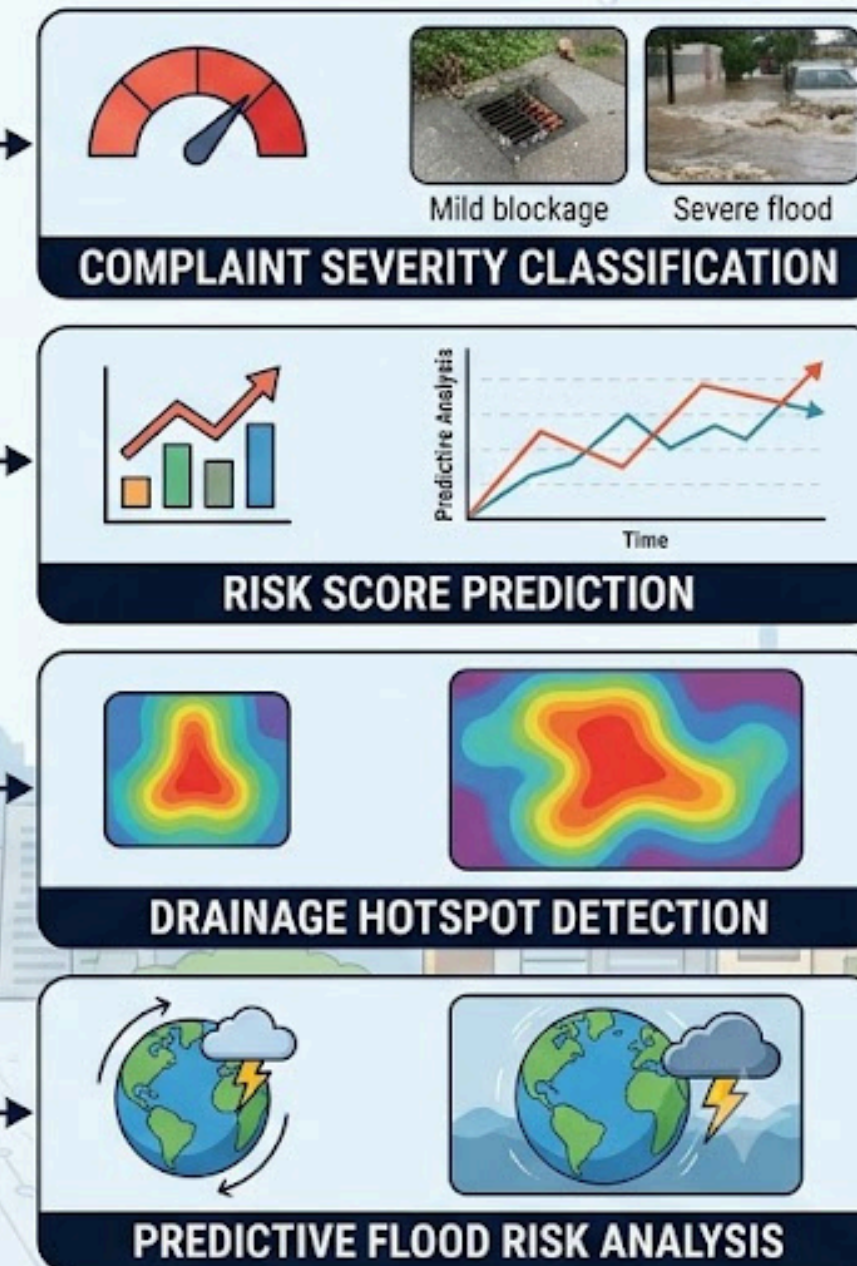
- AI-based complaint prioritization
- GIS-based drainage monitoring
- Real-time complaint tracking
- Predictive hotspot analysis

'DRAINWATCH' TECHNOLOGY STACK AND AI CAPABILITIES

DRAINWATCH TECHNOLOGY STACK



DRAINWATCH AI FEATURES



THANK YOU